

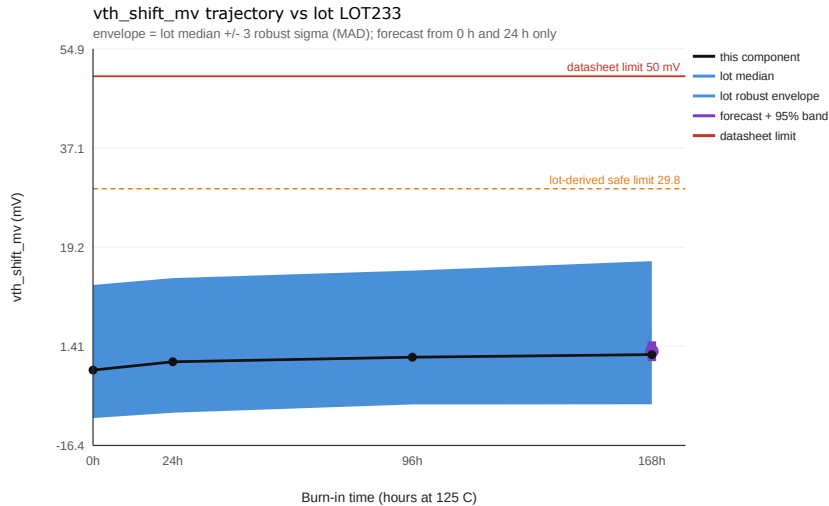
QA DISPOSITION REPORT burn-in parametric screening

|           |              |               |                                                         |       |            |
|-----------|--------------|---------------|---------------------------------------------------------|-------|------------|
| Component | LOT233-C0222 | Lot           | LOT233                                                  | Board | LOT233-B03 |
| Socket    | r3 c6        | Model version | dataset-v1.1 / ModuleA-d13e1ef / ModuleB-260260a2710:40 |       |            |

DECISION: WATCH

LAYER 1 — deterministic evidence (every value cites its comparison basis)

|                    |                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Absolute limit     | PASS (8.73 ns vs 10.0 ns max, prop_delay_ns)                                                                                          |
| Lot-relative level | +0.9 robust sigma vs lot median on leakage_na (median 26.46 nA, sigma 5.59 nA, n=499, MAD estimator; AEC-Q001 IQR/1.35 gives 5.70 nA) |
| Drift 0h to 24h    | +1.483 mV on vth_shift_mv; lot median drift +1.001 mV, part is +0.6 robust sigma on the lot's drift distribution                      |
| Drift rate         | 0.0618 mV/h (lot median 0.0417 mV/h, 1.5x)                                                                                            |
| Max interval jump  | +0.7 robust sigma on iddq_ua; largest interval change is 3.5x the median of the others (concentrated -> step)                         |
| Predicted 168h     | 0.522 mV (vth_shift_mv, from 0h and 24h only)                                                                                         |
| 95% upper bound    | 2.303 mV -> within lot-derived safe limit 29.754 mV                                                                                   |
| Joint anomaly      | Mahalanobis D2 = 23.0 (good-part 99th pct = 19.7); largest contribution leakage_na at 36% of the total                                |
| Leakage/delay sign | leakage +0.9 sigma, delay +0.7 sigma -- consistent with the lot's process locus                                                       |
| Spatial check      | no neighbour clustering on board LOT233-B03 (2/64 flagged), component-specific                                                        |
| Data quality       | all checkpoints valid                                                                                                                 |
| Primary reason     | joint-distribution anomaly across parameters (univariately normal)                                                                    |
| Action             | Release, but tag the component ID for trend monitoring at the next screen. No rework.                                                 |



LAYER 2 — model attribution

| detector       | normalised strength | fired | feature (Module B) | SHAP (mV) |
|----------------|---------------------|-------|--------------------|-----------|
| L3a_MCD        | 1.34                | YES   | v24                | -2.285    |
| L3b_Q          | 0.93                | no    | z:d1               | -0.119    |
| L4d_AutoEnc    | 0.19                | no    | s1                 | +0.116    |
| L4b_LOF        | -0.02               | no    | d1                 | +0.082    |
| L1_dynamic_MAD | -0.15               | no    | lotmed_t0          | -0.077    |

Detector strength is distance past that detector's own operating threshold (1.0 = exactly at it). SHAP values are exact TreeSHAP from the LightGBM drift model, in target units. **Caveat:** v0 and v24 are 0.99 correlated, so SHAP credit between them is not identifiable — a Huber model of equal accuracy attributes it the other way round. Treat Layer 2 as supporting evidence; Layer 1 is the auditable record.

COUNTERFACTUAL

This part would have moved from WATCH to PASS if iddq\_ua at 96 h had been at or above 11.499 uA (measured 11.373 uA, a change of 0.126 uA = 0.0 robust sigma).

LAYER 3 — failure-mechanism HYPOTHESIS (for the FA engineer; not a diagnosis)

**localised defect not yet affecting timing paths; the class that escapes univariate screening** [confidence: speculative]  
*Because:* every parameter is within its normal range, but the joint distance is  $D^2=23.0$  against a good-part 99th percentile of 19.7  
*Caveat:* This is the weakest mapping in the table and should be labelled as such. A joint-distribution outlier says the part does not resemble its peers; it does NOT say why. The value is in flagging the part for attention, not in naming the mechanism.  
*Suggested next test:* quarantine and prioritise for FA; treat the mechanism as unknown

**SIGN-OFF**

|             |  |           |  |
|-------------|--|-----------|--|
| Inspector   |  | Date      |  |
| Disposition |  | Signature |  |
| Notes       |  |           |  |

Decision support only. Final rejection remains under approved engineering rules and QA control; this system detects, ranks, explains and forecasts. The model never overrides a hard engineering limit in either direction.